

October this year. Each station would require around 300 square metres to generate 20kW power." "There are many water bodies that could be used for solar power projects," he says.

What is leading to this growth

Growth needs driving factors. Government of India is playing a major role in growth of the solar sector today. The Central government allows substantial concession or exemption of customs duty on import of equipment required to set up power projects.

Then there is rising demand for electricity, favourable government policies and high power tariffs that support the growth of solar industry.



Fig. 2: Key growth drivers and inhibitors for solar energy market in India (Source: Novonous Insight)

The declaration of National Solar Mission (NSM) a few years ago has increased the hope of solar module manufacturers for growth in gross revenue. This has flooded the Indian market with cheaper imported modules and equipment.

One important attempt has been the release of funds by the Indian government to support the industry. S.K. Sharma, scientist D, Ministry of New and Renewable Energy (MNRE), in one of his reports says, "I am directed to convey sanction of the President for installation of 23,500 nos. (of) SPV power plants of 1kW capacity each for domestic households in Rajasthan, Andhra Pradesh, Kerala, Tamilnadu, Chhattisgarh and other selected States with a financial support from NCEF of ₹ 148 crores (₹ 1480 million), in Rajasthan, Andhra Pradesh, Kerala, Tamil Nadu, Chhattisgarh and other selected States at a project cost of ₹ 493 crores (₹ 4930 million)."

Another factor supporting the growth is the heavy import duty imposed on the Chinese modules in the USA. This is acting as a motivational factor for the solar manufacturers in India to make their way into the lucrative US market.

The contribution of manufacturers in the solar industry has also been a big driving factor for the development of the solar electronics and the industry as a whole. For instance, Honeywell has

recently added its sixth manufacturing facility at Vadodra. It is stated that this unit will construct ten solar water-heater variants of galvanised iron and stainless steel (SS) that would deliver 100 to 300 litres per day for industrial, commercial and residential use. Then there is Schneider Electric's Bengaluru plant, which has a capability to manufacture solar inverters that can raise nearly 1.4GW of energy a year.

Addition to the Indian solar team

The market is not a place for a single player. There are always new players that add up to the existing team to make the market grow. Solar industry in India needs many such players to evolve. Canadian Solar, a Chinese company and the world's third-largest photovoltaic cell manufacturer, is aiming to set up its manufacturing base in India. But this is not all, it is also likely to take over Solar Semiconductor based in Hyderabad which is capable of producing about 195MW worth of modules. Then there are other solar companies like CNBN, Yingli, Trina and ReneSola from China that are also trying to set up their manufacturing base in India.

Su-Kam Power Systems has joined hands with Dhampur Mills to sell solar products in the rural areas of Uttarakhand and Uttar Pradesh (UP) through 'E-HAATS'. Dhananjay Sharma, general manager-solar networks, Su-Kam in one of his interview to *Electronics Bazaar* says, "Through this initiative, Su-Kam will sell technologically advanced solar products that offer people value for money, thereby helping them in both generating and saving energy through solar power usage."

Indian market trend

Indian solar industry has built up quite interesting market trends through the solar power systems and other solar electronics products. Be it the Gujarat model or the new government schemes, all of them have acted as the pillars for growth of the solar market in India. Another pillar is the growing technology, where a portable, solar-powered mobile

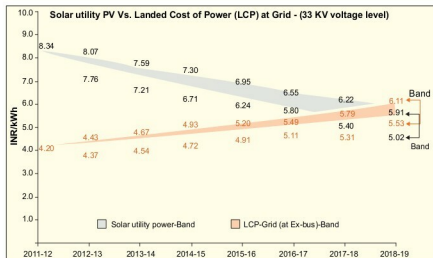


Fig. 3: Grid parity for solar power—utility level (Source: KPMG's solar grid parity model)